

Capstone Project

Nova Scotia Health & Population Analytics

Team Alpha

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Year

☐ Select all

☐ 2022

AgeBand

☐ Select all

☐ Children

☐ Senior

☐ Working Age

Gender

☐ Select all

☐ F

☐ M



Key Highlight
In 2022, Nova Scotia recorded a **chronic disease prevalence rate** of 147 per 1,000 residents — reflecting a significant and growing burden on the provincial health system. **Hypertension** emerged as the leading condition, highlighting a largely preventable disease as the primary driver of chronic illness across the province. **The Eastern health zone** carries the highest risk among all regions, while **adults aged 80 and over** represent the most affected population cohort.

- These findings point to an urgent need for targeted prevention strategies, equitable resource allocation to high-risk zones, and strengthened support for Nova Scotia's aging population.

How Has Nova Scotia Population Shifted Over The Last Decade & What Are The Projected Trends

Year

Select all

2012

2013

2014

2015

2016

2017

2018

AgeBand

Select all

Children

Senior

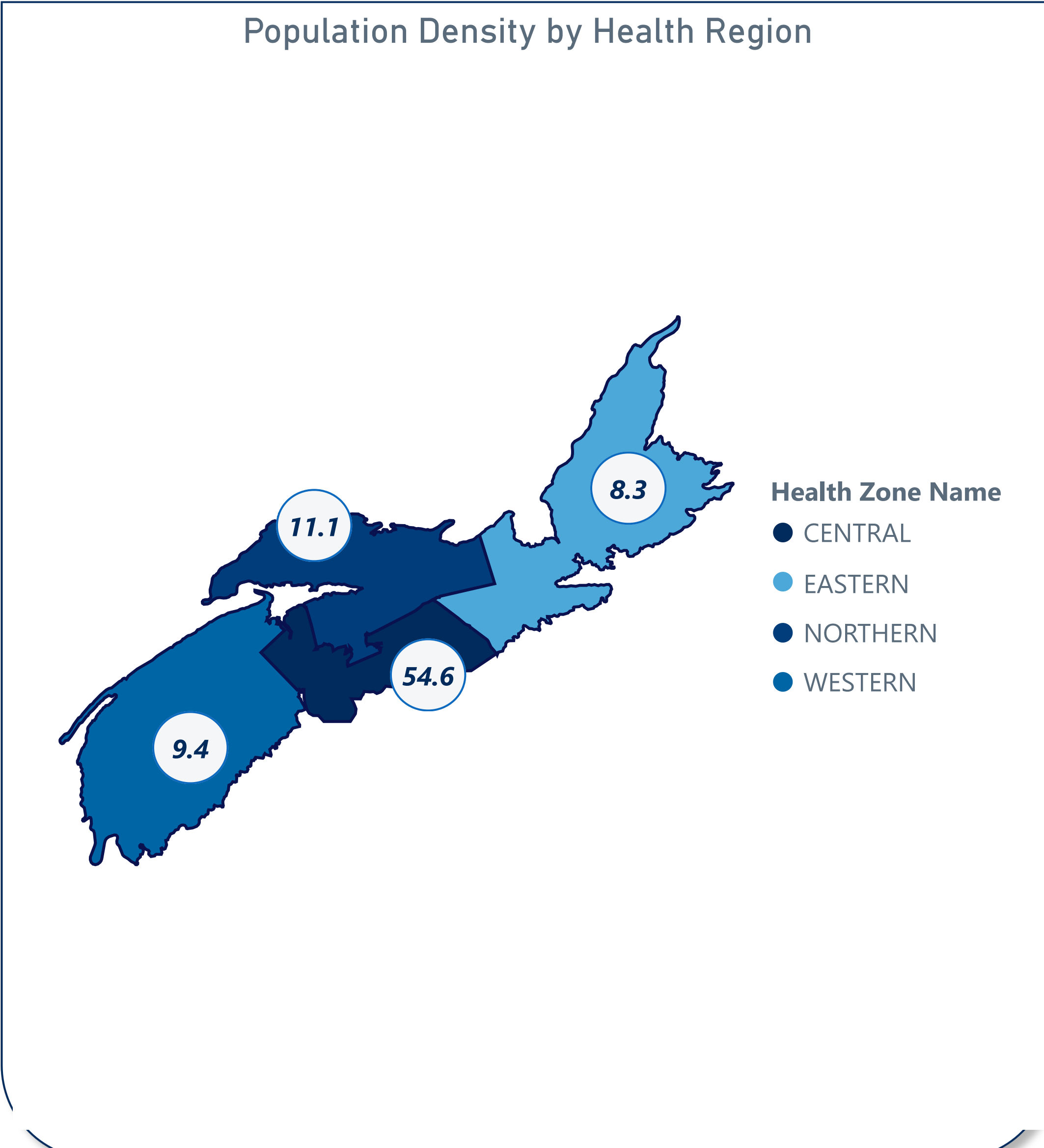
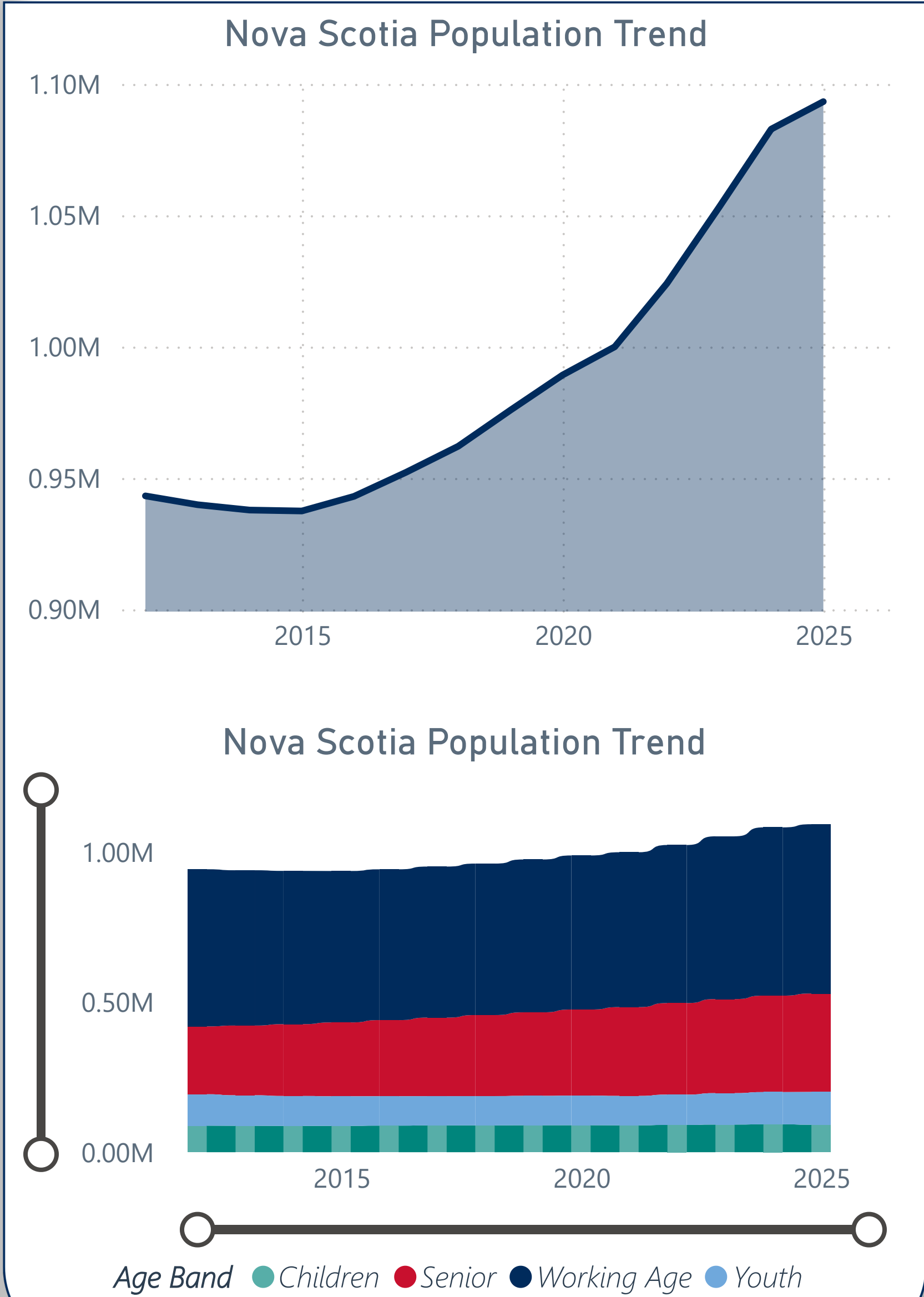
Working Age

Gender+

Select all

Men

Women



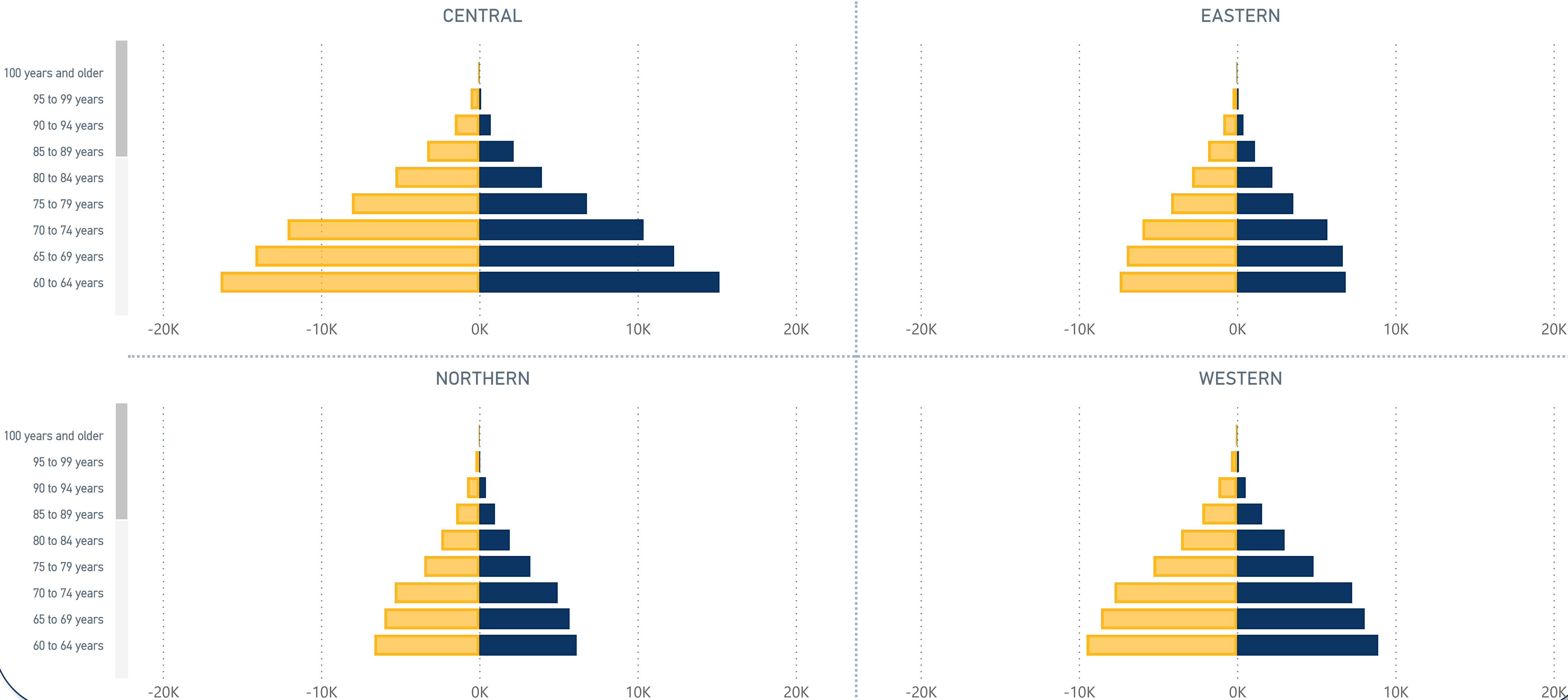


Nova Scotia Health Zone Demographic Breakdown

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Demographic Composition Across Nova Scotia Health Regions(2021)

● Female Population ● Male Population





What NS Health Zones Show The Highest Rates of Chronic Diseases

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Chronic Disease

- ☐ Select all
- ☐ Acute Myocardial Infarction ...
- ☐ Asthma
- ☐ Chronic Obstructive Pulmon...
- ☐ Diabetes
- ☐ Hypertension
- ☐ Ischemic Heart Disease

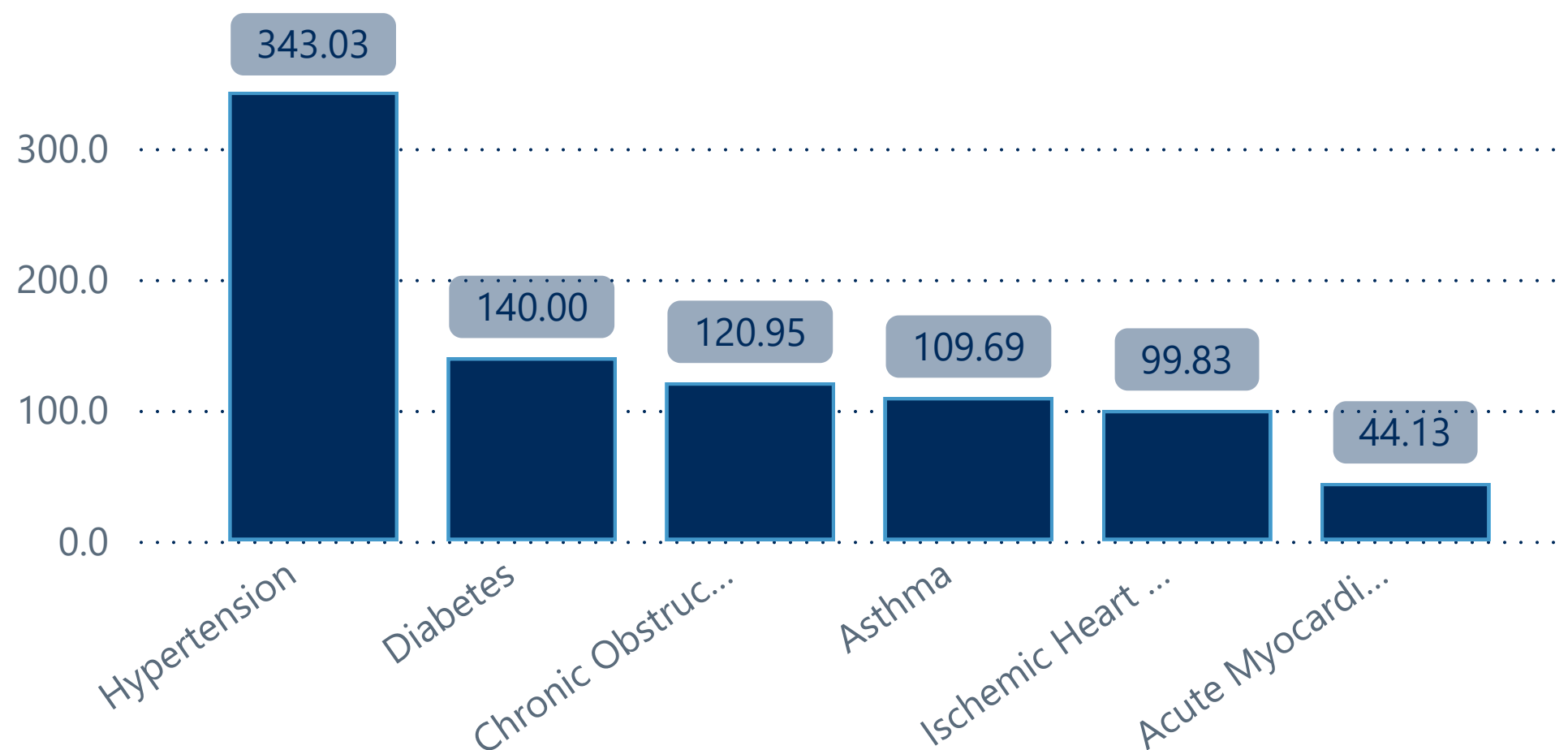
Sex

- ☐ Select all
- ☐ F
- ☐ M

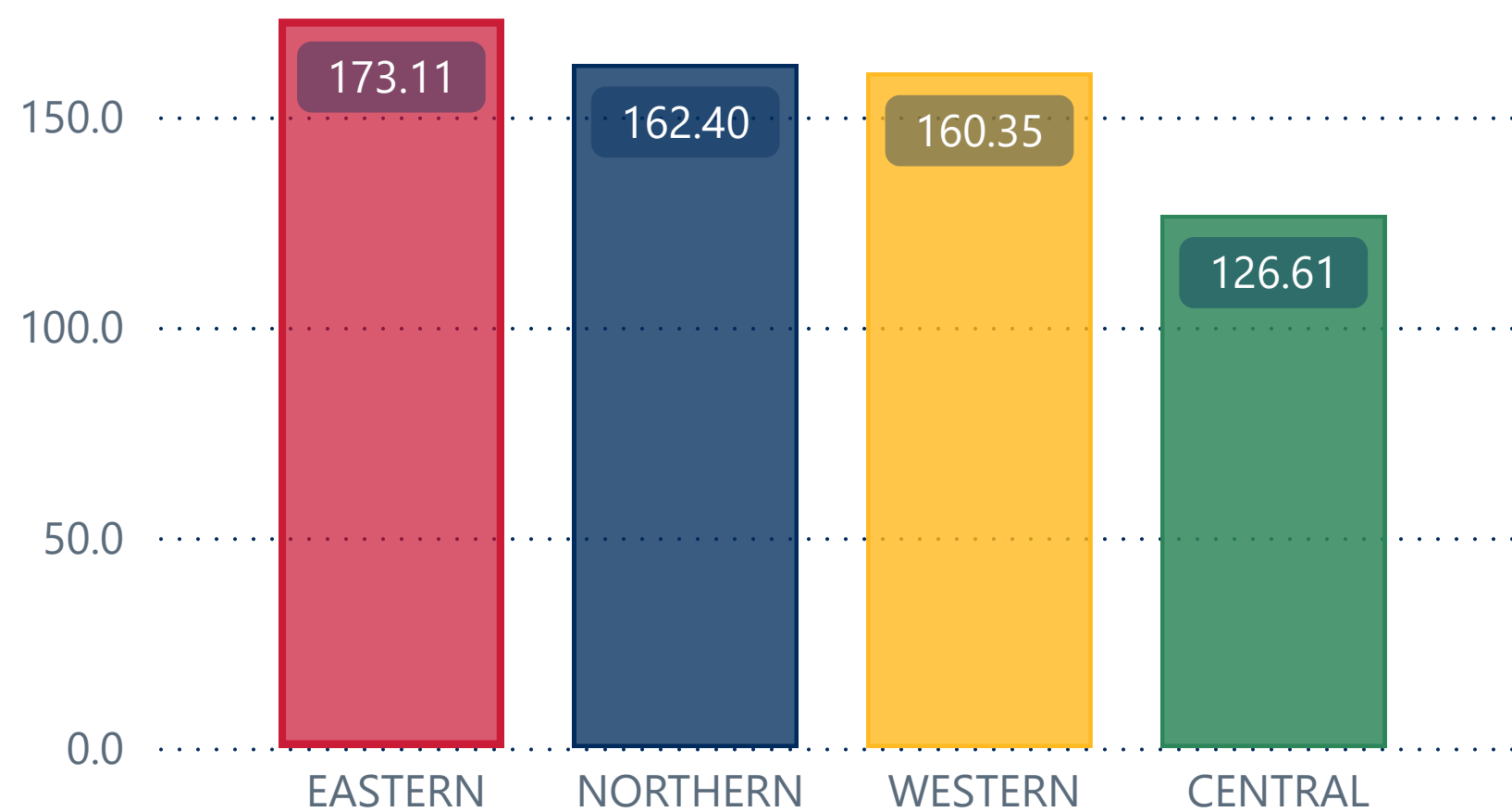
Sex

- ☐ Select all
- ☐ CENTRAL
- ☐ EASTERN
- ☐ NORTHERN
- ☐ WESTERN

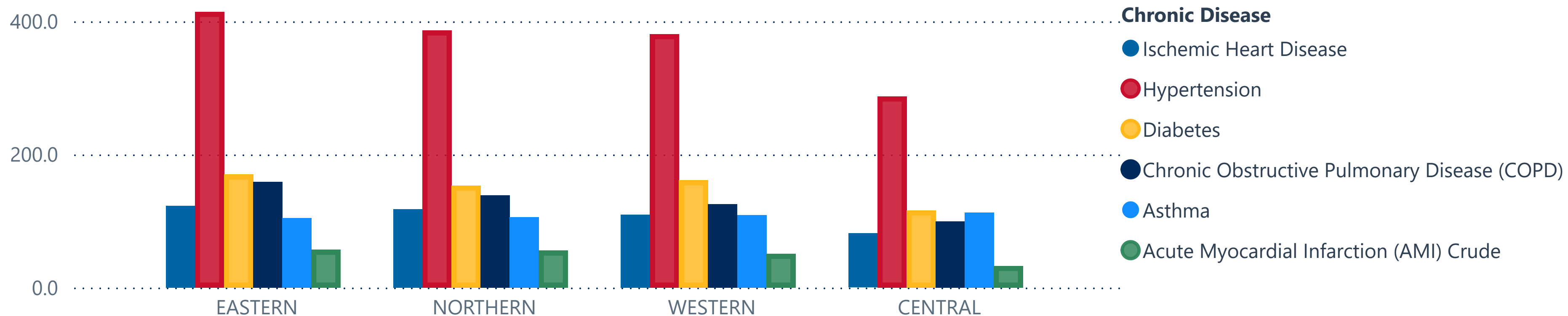
NS Chronic Disease Prevalance (2022)



Chronic Disease Burden Across NS Health Zones



Chronic Disease Prevalence by Zone (per 1000)



Who Are Driving HealthCare Demand?

Year

- ☐ Select all
- ☐ 2012
- ☐ 2013
- ☐ 2014
- ☐ 2015
- ☐ 2016

Health Zone Name

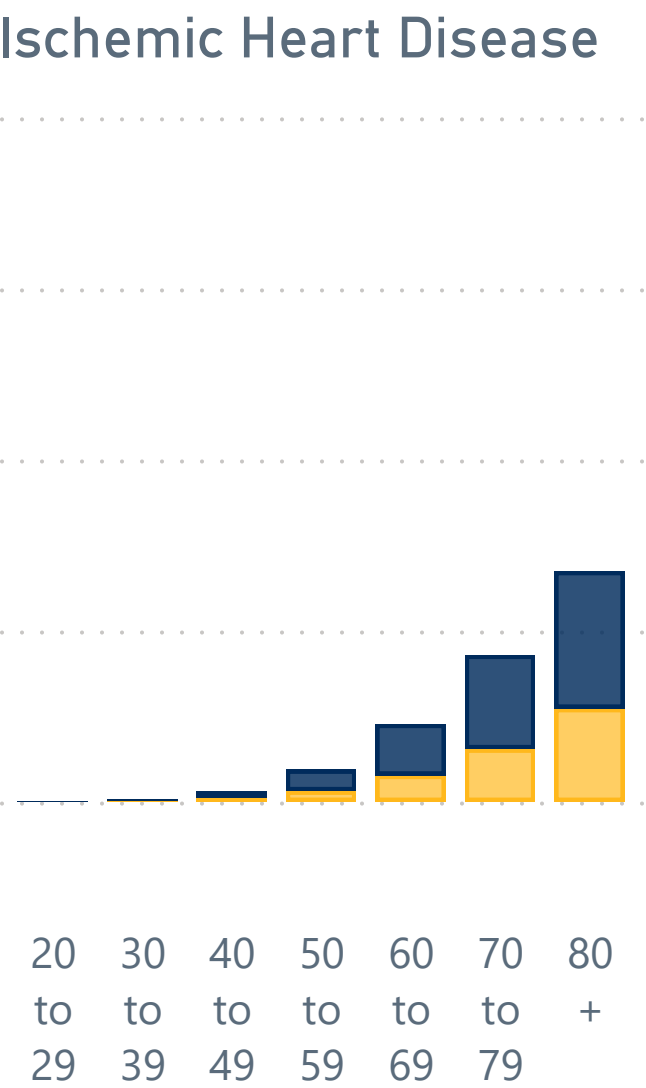
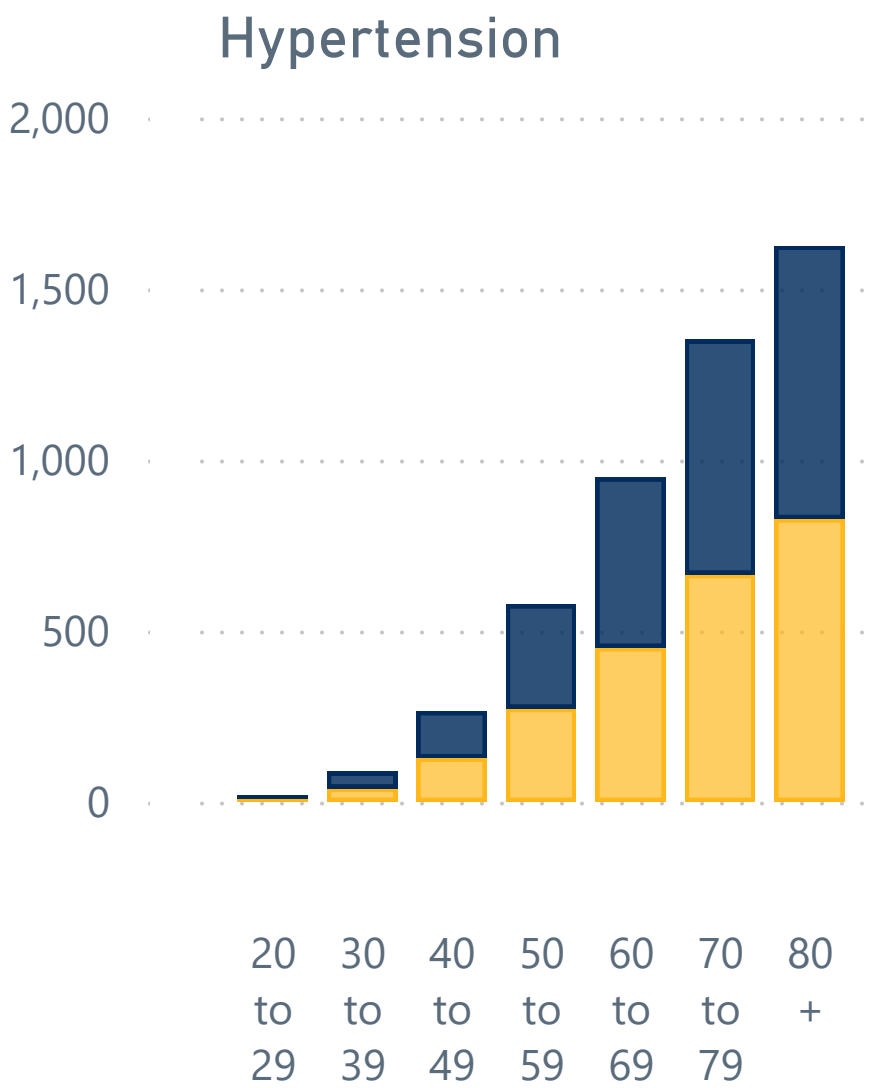
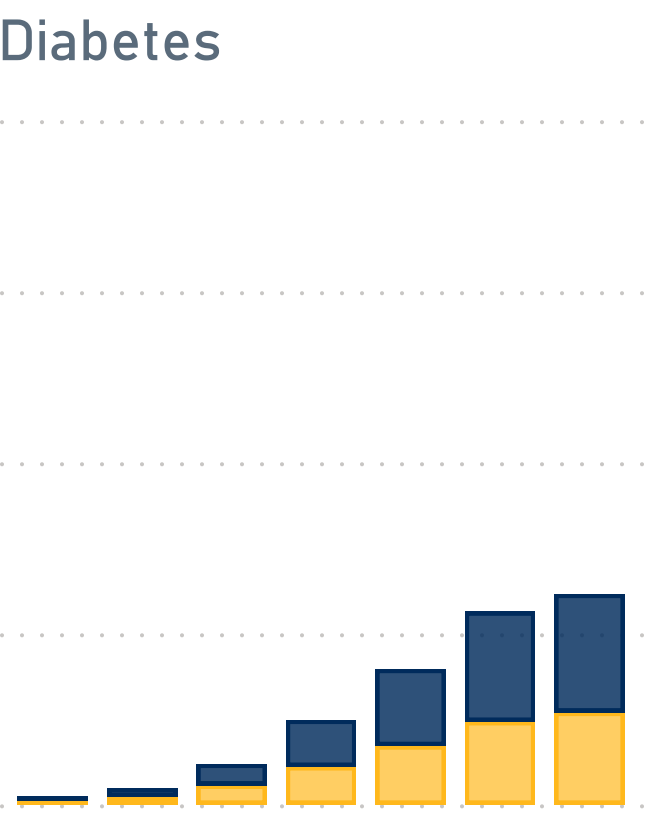
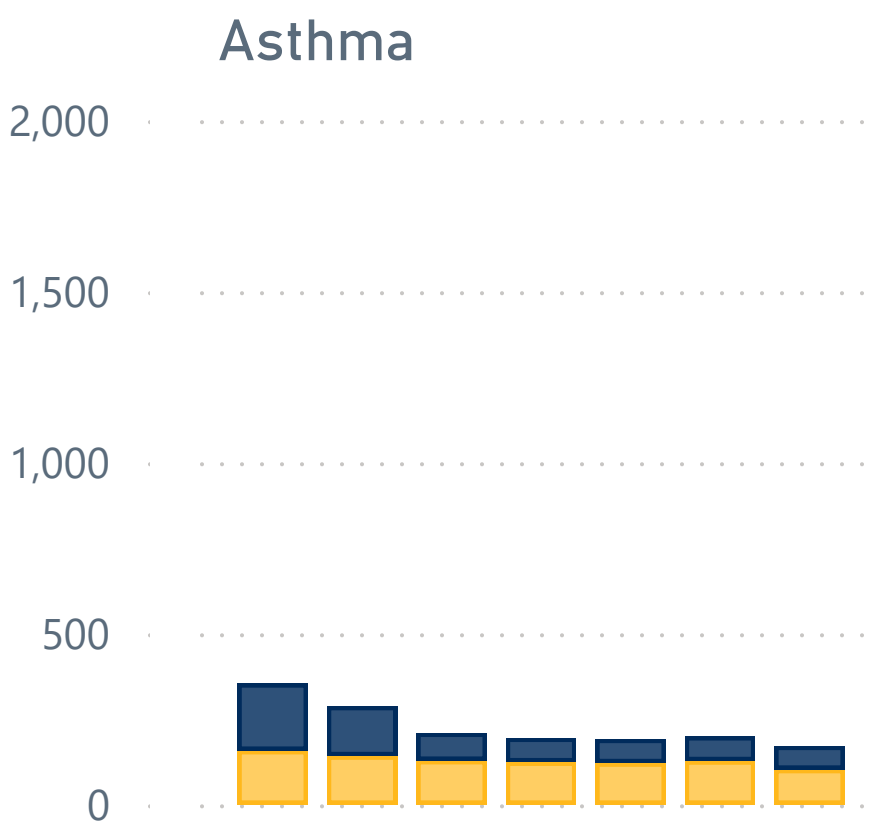
- ☐ Select all
- ☐ CENTRAL
- ☐ EASTERN

Sex

- ☐ Select all
- ☐ All
- ☐ F
- ☐ M

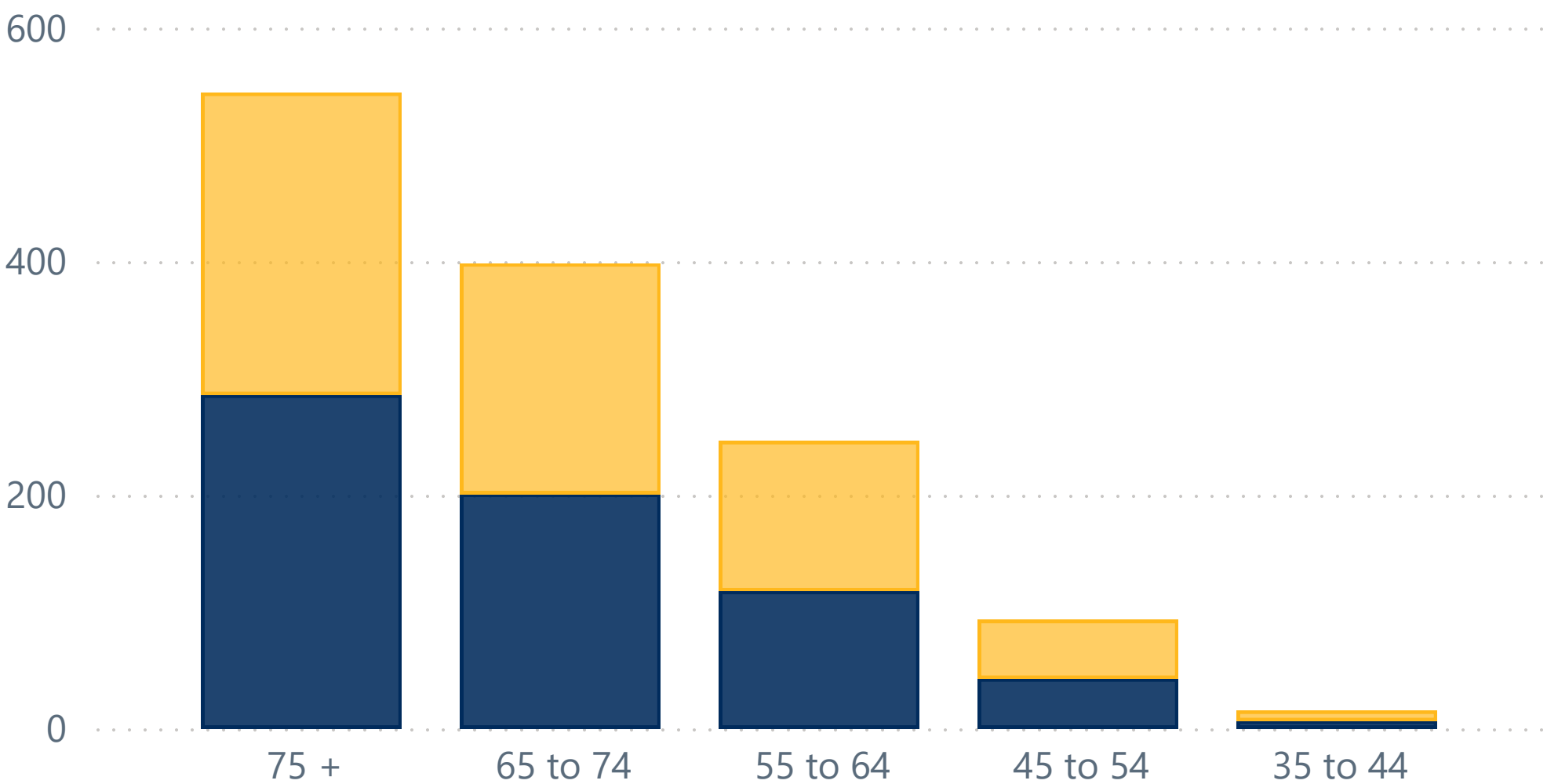
Prevalence Rate per 1000 by Age and Gender

F M

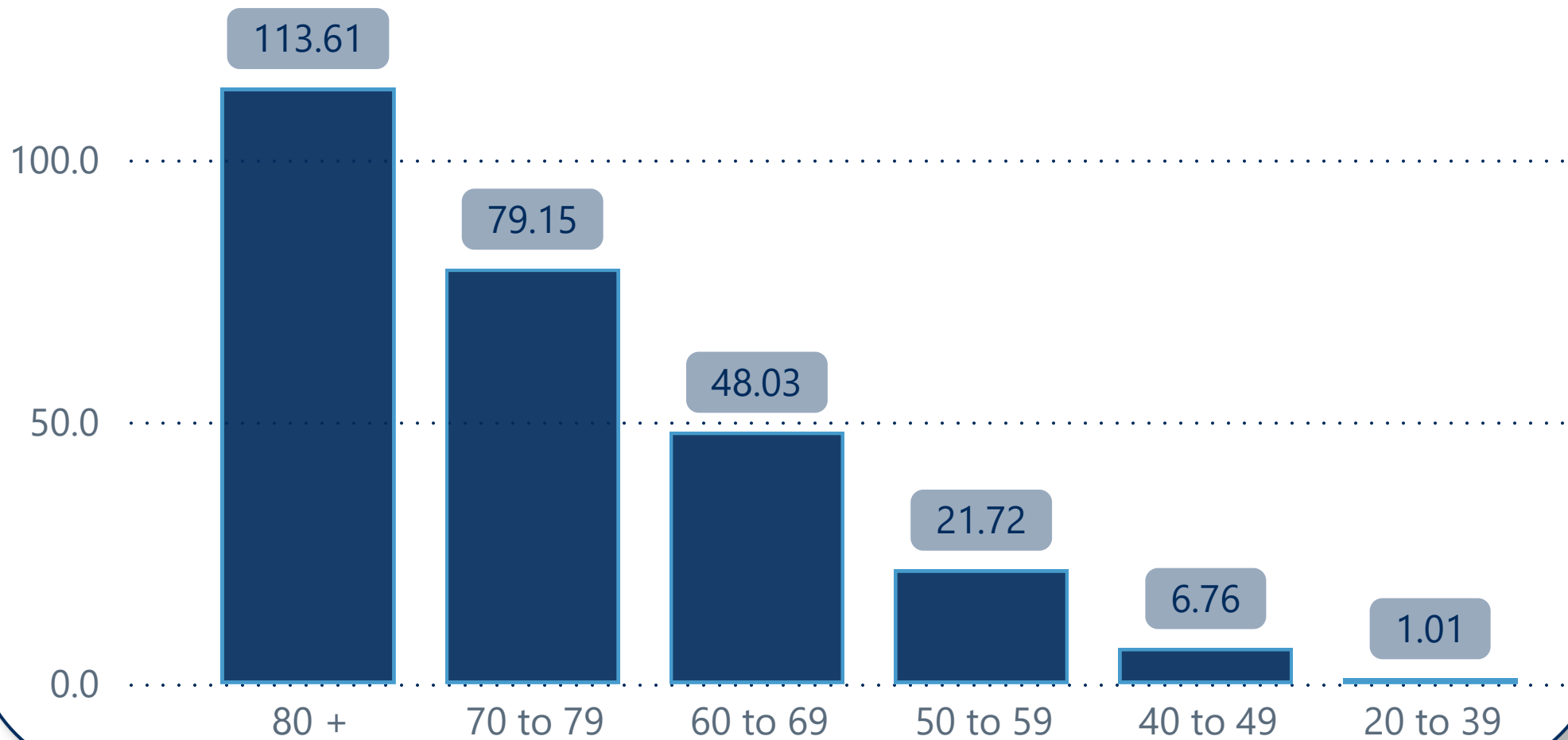


COPD Prevalence Rate per 1000 by Age and Gender

M F



AMI Prevalence by Age





Is There A Correlation Between Income and Education & Chronic Diseases In NS

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Relationship Between Income & Education and Chronic Diseases

Socioeconomic_Label ▼	AMI	Asthma	COPD	Diabetes	Disease Burden (Avg)	Hypertension	IHD	Life Expectancy
Upper Secondary %	-0.67	0.19	0.93	-0.59	0.74	0.91	0.90	0.64
Tertiary Edu %	0.87	0.11	-0.83	0.81	-0.54	-0.81	-0.97	-0.47
Median Income	0.93	0.56	-0.45	0.88	-0.10	-0.49	-0.78	0.02
Below Upper Sec %	-0.98	-0.47	0.58	-0.96	0.20	0.57	0.91	0.18
Average Income	0.89	0.41	-0.56	0.82	-0.25	-0.60	-0.82	-0.10

Year

☐ Select all

☐ 2012

☐ 2013

☐ 2014

☐ 2015

☐ 2016

☐ 2017

☐ 2018

☐ 2019

☐ 2020

☐ 2021

☐ 2022

GeoLevel

☐ Select all

☐ National

☐ Province

Corr Hypertension vs Upper Secondary

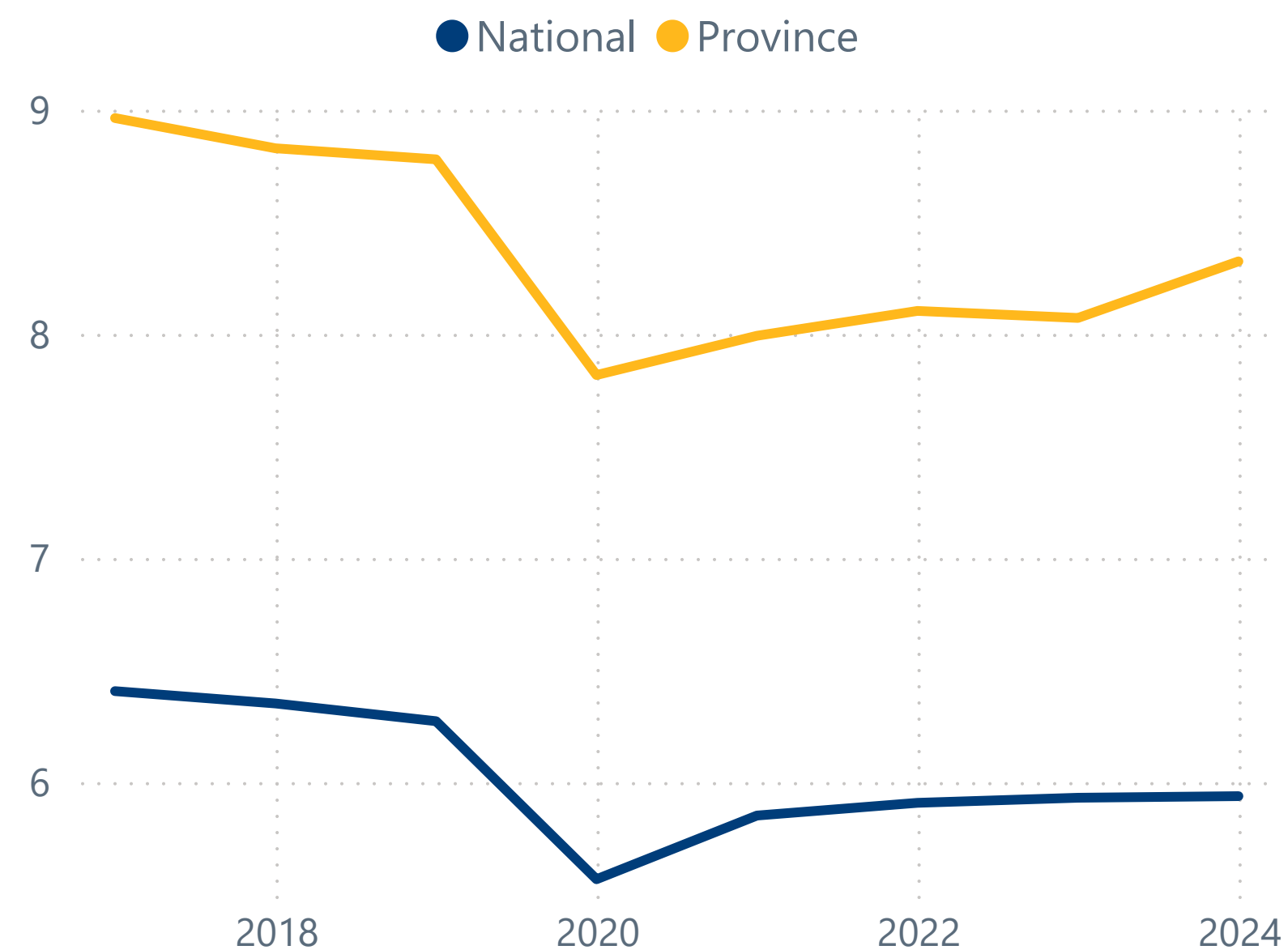
0.91

Corr Hypertension vs Tertiary Edu

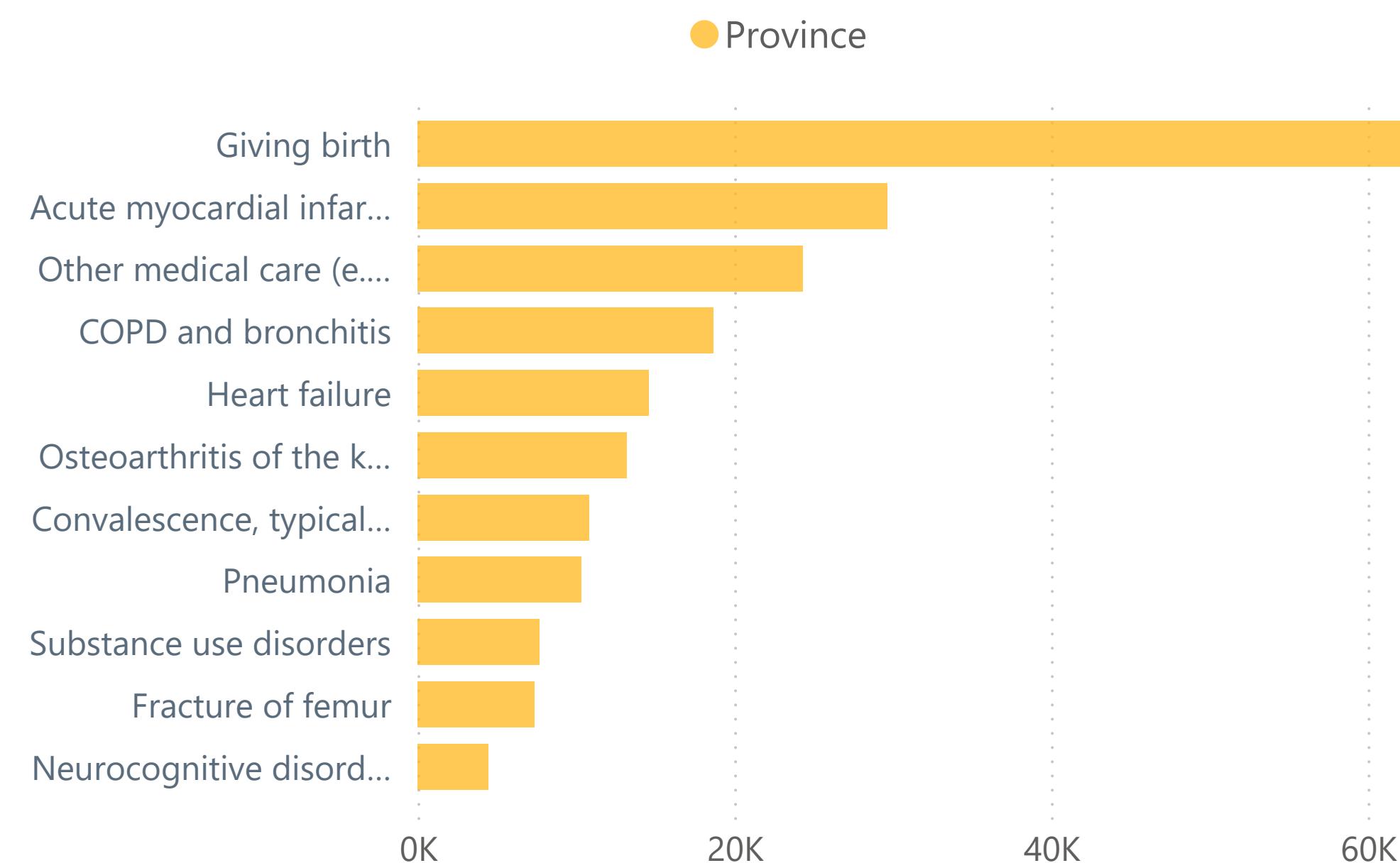
-0.81

Nova Scotia Hospitalization Compared To The National Average

NS hospitalization rates compared to the national average



Most Responsible Diagnosis For inp Hospitalization





Nova Scotia Health & Population Analytics

Chronic Disease Projection 2023 - 2030

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Year

- ☐ Select all
- ☐ 2012
- ☐ 2013
- ☐ 2014
- ☐ 2015

Health Zone Name

- ☐ Select all
- ☐ CENTRAL
- ☐ EASTERN

Chronic Disease

- ☐ Select all
- ☐ Acute Myocardial Infarc...
- ☐ Asthma

Sex

- ☐ Select all
- ☐ All
- ☐ F
- ☐ M

Predict prevalence rate per 1000

172
2030

Most Prevalent Disease

Hypertension

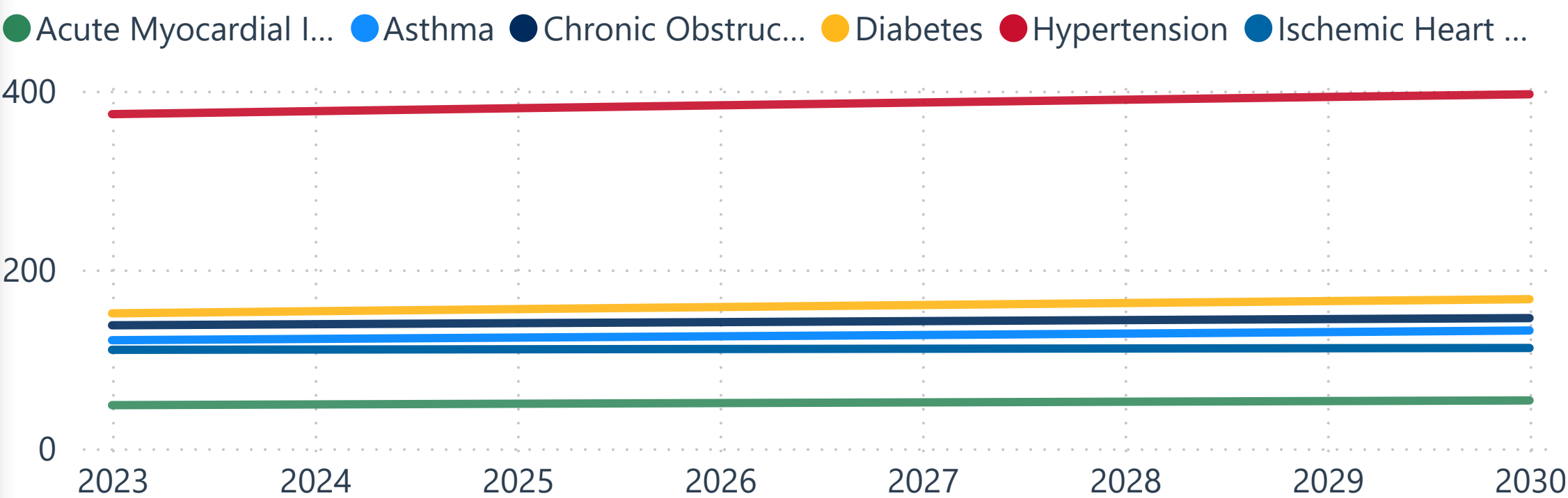
Most Affected Age Cohort

80 +

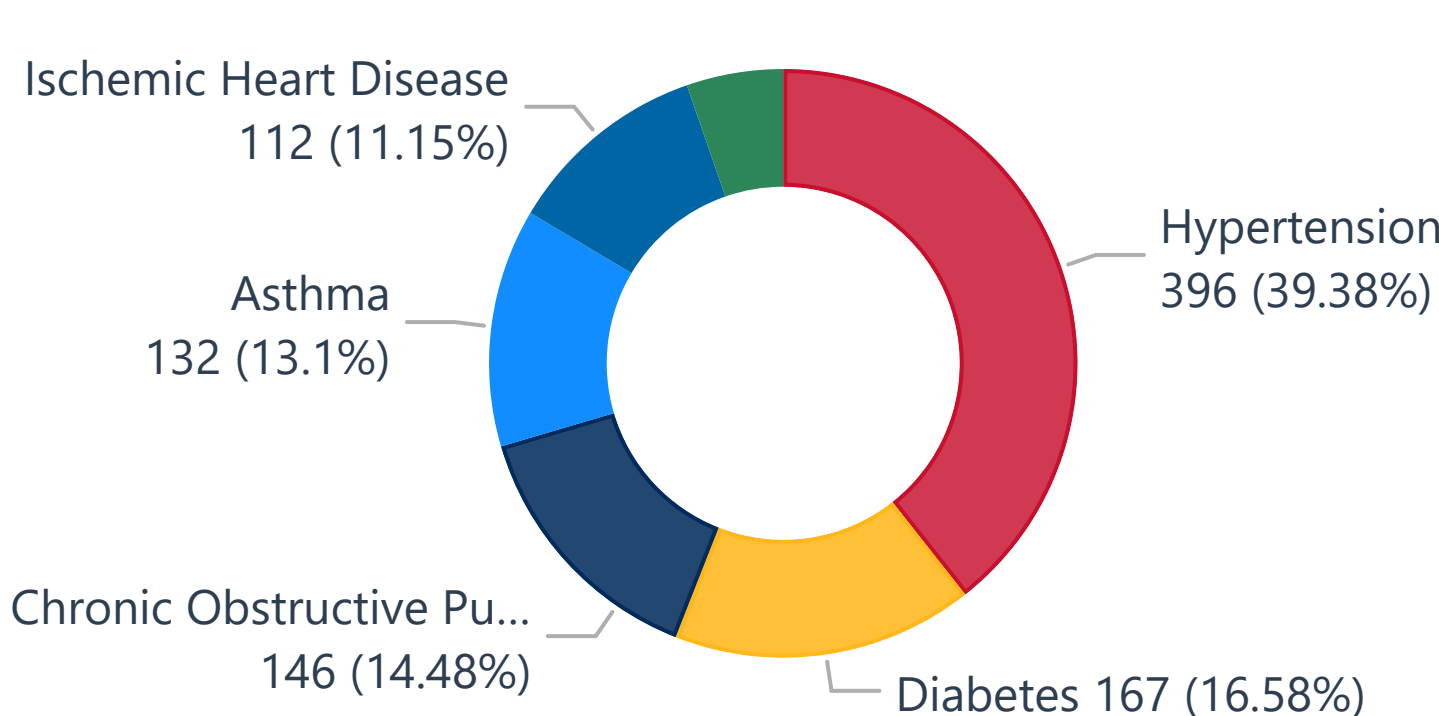
Highest Risk Zone

EASTERN

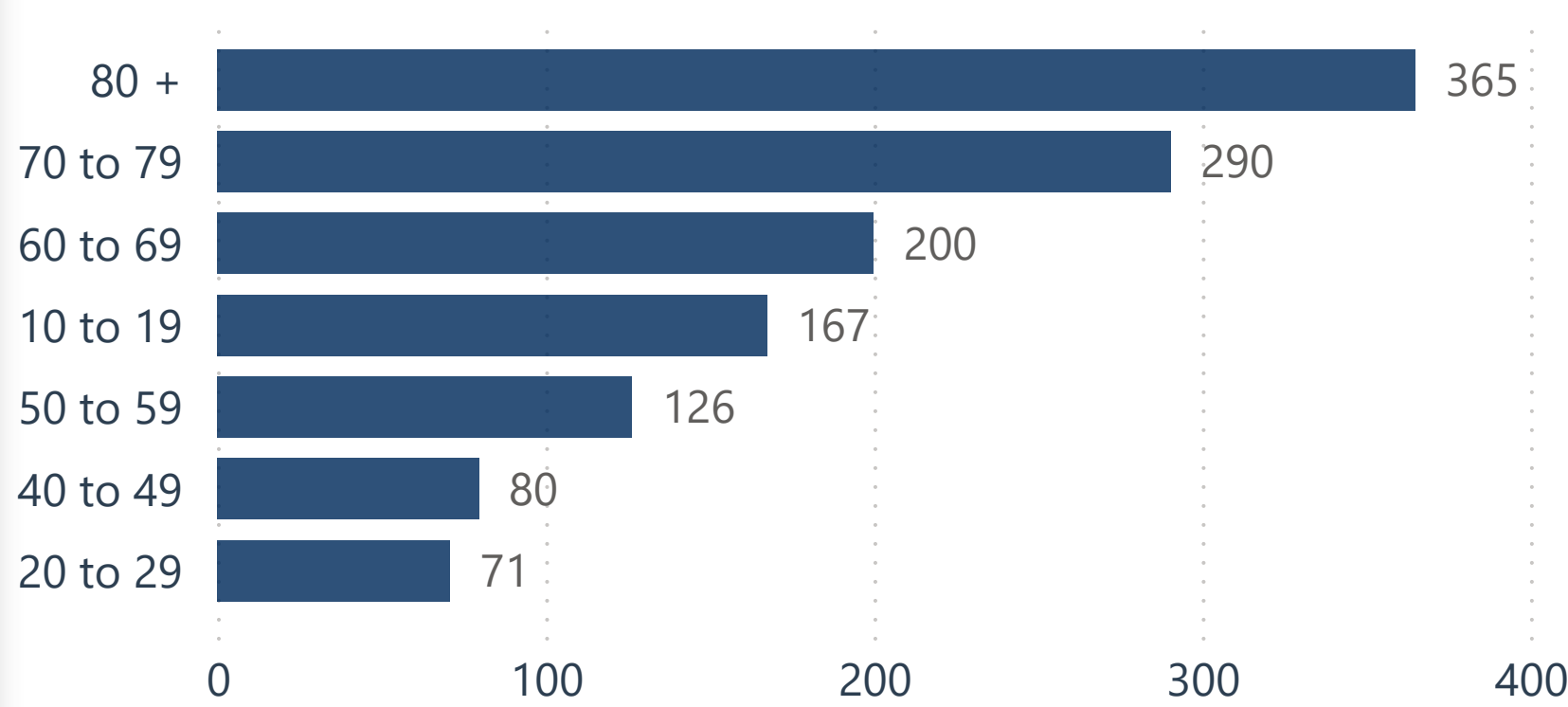
Predict Prevalence Rate per 1000 by Year and Chronic Disease



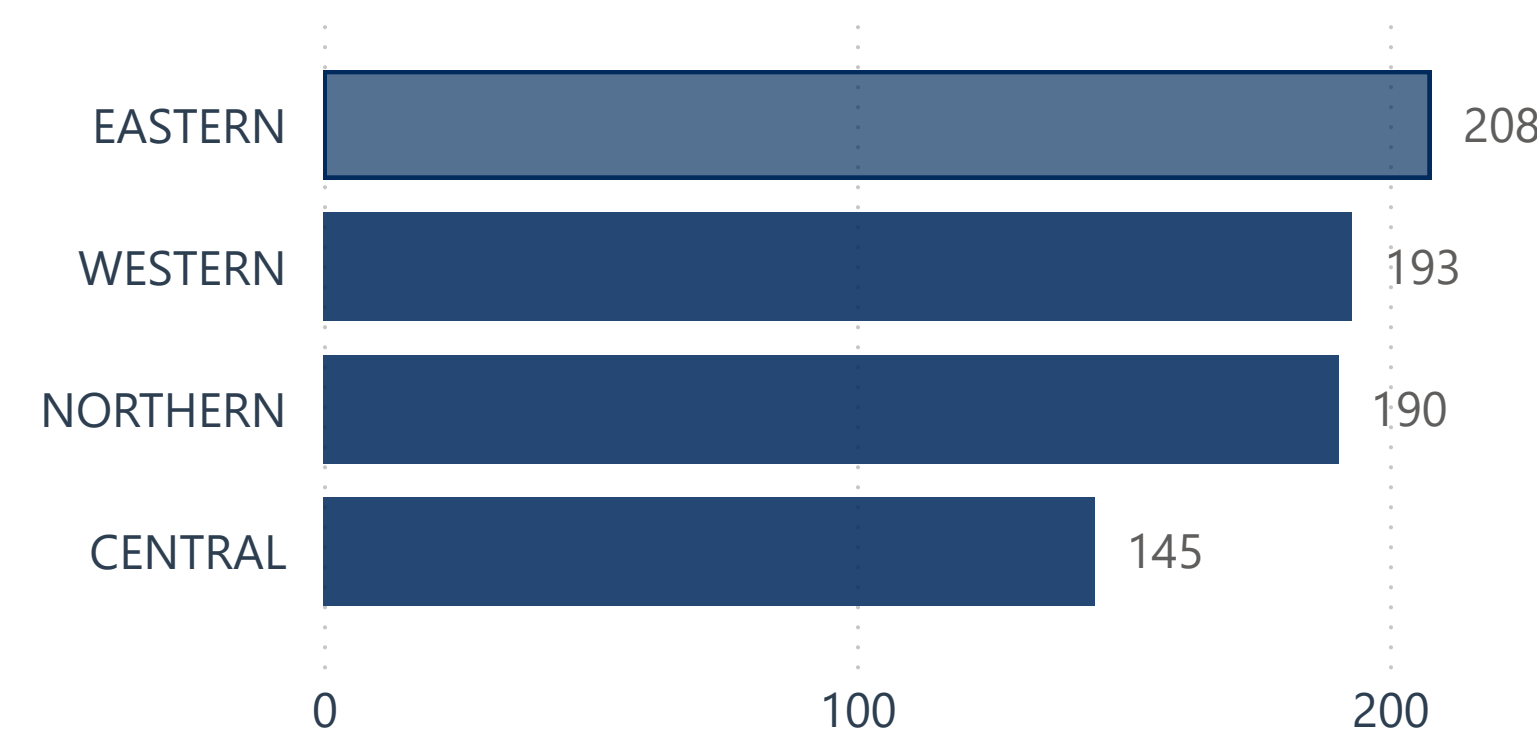
Projected prevalence 2030 by Chronic Disease



Projected prevalence 2030 by CommonBand



Predict Prevalence Rate per 1000 by Health Zone Name



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Recommendations

Drive Targeted Hypertension Awareness Campaigns Across Nova Scotia Focus on lifestyle modification, early detection, consistent screening, and prevention — particularly in zones with the highest burden (Eastern and Northern). With hypertension prevalence at 343 per 1,000 and a strong correlation with lower education attainment (0.91), campaigns should prioritize health literacy, simplified self-management resources, and community-level outreach tailored to populations with limited tertiary education.

Deploy Mobile Clinics for Targeted Screening in Aging Communities Prioritize mobile clinic deployment in communities with older age profiles, where the 70+ and 80+ cohorts are driving the sharpest prevalence spikes across hypertension, COPD, ischemic heart disease, and AMI. Mobile units would bring screening directly to underserved rural areas — particularly in the Eastern zone — reducing the gap between disease burden and primary care access.

Increase Access by Extending Specialized Care to High-Burden Zones Decentralize specialized chronic disease management by deploying cardiologists, respiratory specialists, and diabetes educators into the Eastern and Northern zones rather than concentrating them in urban centres. With NS hospitalization rates running 30–50% above the national average and chronic conditions dominating inpatient diagnoses, shifting specialist capacity upstream would intercept patients before they progress to costly acute admissions like AMI



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